Inequalities — Solutions

Problem 1. Let n be a positive integer. Suppose each of $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ is a permutation of $1, \frac{1}{2}, \cdots, \frac{1}{n}$ such that

$$a_1 + b_1 \geqslant a_2 + b_2 \geqslant \cdots \geqslant a_n + b_n.$$

Prove that $a_k + b_k \leqslant \frac{4}{k}$ for each $k = 1, 2, \ldots, n$.

Solution. Suppose on the contrary that $a_k + b_k > \frac{4}{k}$ for some k. By the given condition, we have $a_j + b_j > \frac{4}{k}$ for each $j = 1, 2, \ldots, k$. Therefore, at least one of a_j and b_j is larger than $\frac{2}{k}$. Since there are k choices of j, there are at least k numbers among $a_1, a_2, \ldots, a_k, b_1, b_2, \ldots, b_k$ which are larger than $\frac{2}{k}$.

On the other hand, note that $\frac{1}{j} > \frac{2}{k}$ iff $j < \frac{k}{2}$, or $j \leqslant \left\lfloor \frac{k-1}{2} \right\rfloor$. Thus, there are only $2\left\lfloor \frac{k-1}{2} \right\rfloor$ numbers among $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ which are larger than $\frac{2}{k}$. But then

$$2\left\lfloor \frac{k-1}{2} \right\rfloor \leqslant 2 \cdot \frac{k-1}{2} = k - 1 < k,$$

which is a contradiction. $\square$

Problem 2. (Tournament of Towns 2021 Fall Junior A Level Problem 6) Prove that for any positive integers $a_1, a_2, \ldots, a_n$ the following inequality holds true:

$$\left\lfloor \frac{a_1^2}{a_2} \right\rfloor + \left\lfloor \frac{a_2^2}{a_3} \right\rfloor + \cdots + \left\lfloor \frac{a_n^2}{a_1} \right\rfloor \geqslant a_1 + a_2 + \cdots + a_n.$$

Solution. Let $a_{n+1} = a_1$. For each $j = 1, 2, \ldots, n$, we have

$$\frac{a_j^2}{a_{j+1}} \geqslant 2a_j - a_{j+1}$$

since this is equivalent to $(a_j - a_{j+1})^2 \geqslant 0$. As the right-hand side is a positive integer, we must have

$$\left\lfloor \frac{a_j^2}{a_{j+1}} \right\rfloor \geqslant 2a_j - a_{j+1}.$$

Adding up all these inequalities, we deduce

$$\sum_{j=1}^{n} \left\lfloor \frac{a_j^2}{a_{j+1}} \right\rfloor \geqslant \sum_{j=1}^{n} 2a_j - \sum_{j=1}^{n} a_{j+1} = \sum_{j=1}^{n} a_j$$

as desired. $\square$

Problem 3. Let $x_1, x_2, \ldots, x_n$ be real numbers satisfying $0 \leqslant x_j \leqslant 1$ for each j. Prove that

$$(1 - x_1 x_2 \cdots x_n)^m + \big(1 - (1 - x_1)^m\big)\big(1 - (1 - x_2)^m\big) \cdots \big(1 - (1 - x_n)^m\big) \geqslant 1$$

for any positive integer m.

Solution. Consider an $m \times n$ table. For each cell in the ith row and the jth column, we fill in the number 0 with probability x_j, and fill in the number 1 otherwise (with probability $1 - x_j$).

Now, for each of the m rows, $x_1 x_2 \cdots x_n$ is the probability that all numbers in this row are 0. Thus, $1 - x_1 x_2 \cdots x_n$ is the probability that at least one number in this row is 1. It follows that

$$(1 - x_1 x_2 \cdots x_n)^m$$

is the probability that each row consists of at least one copy of 1.

Next, for the jth column, $(1 - x_j)^m$ is the probability that all numbers in this column are 1. Thus, $1 - (1 - x_j)^m$ is the probability that at least one number in this column is 0. It follows that

$$\big(1 - (1 - x_1)^m\big)\big(1 - (1 - x_2)^m\big) \cdots \big(1 - (1 - x_n)^m\big)$$

is the probability that each column consists of at least one copy of 0.

It is clear that at least one of the following must hold:

- each row consists of at least one copy of 1;
- each column consists of at least one copy of 0.

Therefore, the inequality

$$(1 - x_1 x_2 \cdots x_n)^m + \big(1 - (1 - x_1)^m\big)\big(1 - (1 - x_2)^m\big) \cdots \big(1 - (1 - x_n)^m\big) \geqslant 1$$

must hold by using probability. $\square$

Problem 4. (Balkan MO Shortlist 2022 A2) Let $k > 1$ be a real number, $n \geqslant 3$ be an integer, and $x_1 \geqslant x_2 \geqslant \cdots \geqslant x_n$ be positive real numbers. Prove that

$$\frac{x_1 + kx_2}{x_2 + x_3} + \frac{x_2 + kx_3}{x_3 + x_4} + \cdots + \frac{x_n + kx_1}{x_1 + x_2} \geqslant \frac{n(k+1)}{2}.$$

Solution. We first prove

$$\frac{x_2}{x_2 + x_3} + \frac{x_3}{x_3 + x_4} + \cdots + \frac{x_n}{x_n + x_1} + \frac{x_1}{x_1 + x_2} \geqslant \frac{n}{2} \tag{1}$$

by induction. For the base case $n = 3$, we have

$$\begin{aligned}
& \frac{x_2}{x_2+x_3}+\frac{x_3}{x_3+x_1}+\frac{x_1}{x_1+x_2} \geqslant \frac{3}{2} \\
\Leftrightarrow \quad & 2\sum_{\text{cyc}} x_2(x_3+x_1)(x_1+x_2) \geqslant 3(x_2+x_3)(x_3+x_1)(x_1+x_2) \\
\Leftrightarrow \quad & x_1^2x_2+x_2^2x_3+x_3^2x_1 \geqslant x_1x_2^2+x_2x_3^2+x_3x_1^2 \\
\Leftrightarrow \quad & (x_1-x_2)(x_2-x_3)(x_1-x_3) \geqslant 0.
\end{aligned}$$

This holds since $x_1 \geqslant x_2 \geqslant x_3$. For the inductive step, assume

$$\frac{x_2}{x_2+x_3}+\frac{x_3}{x_3+x_4}+\cdots+\frac{x_{n-1}}{x_{n-1}+x_1}+\frac{x_1}{x_1+x_2} \geqslant \frac{n-1}{2}$$

for some $n \geqslant 4$. Then we have

$$\frac{x_2}{x_2+x_3}+\frac{x_3}{x_3+x_4}+\cdots+\frac{x_n}{x_n+x_1}+\frac{x_1}{x_1+x_2} \geqslant \frac{n-1}{2}-\frac{x_{n-1}}{x_{n-1}+x_1}+\frac{x_{n-1}}{x_{n-1}+x_n}+\frac{x_n}{x_n+x_1}.$$

So it suffices to prove

$$\frac{x_{n-1}}{x_{n-1}+x_n}+\frac{x_n}{x_n+x_1} \geqslant \frac{1}{2}+\frac{x_{n-1}}{x_{n-1}+x_1}.$$

Clearing denominators and factorizing, this means $(x_1-x_{n-1})(x_{n-1}-x_n)(x_1-x_n) \geqslant 0$, which holds since $x_1 \geqslant x_{n-1} \geqslant x_n$. Therefore, (1) is proved. Finally, we have

$$\begin{aligned}
& \frac{x_1+kx_2}{x_2+x_3}+\frac{x_2+kx_3}{x_3+x_4}+\cdots+\frac{x_n+kx_1}{x_1+x_2} \\
&= \left(\frac{x_1+x_2}{x_2+x_3}+\frac{x_2+x_3}{x_3+x_4}+\cdots+\frac{x_n+x_1}{x_1+x_2}\right)+(k-1)\left(\frac{x_2}{x_2+x_3}+\frac{x_3}{x_3+x_4}+\cdots+\frac{x_1}{x_1+x_2}\right) \\
&\geqslant n+(k-1)\cdot\frac{n}{2} \quad \text{(by the AM-GM inequality and (1))} \\
&= \frac{n(k+1)}{2}.
\end{aligned}$$

□

Problem 5. (China TST 2018 Test 1 Problem 6) Let $A_1, A_2, \ldots, A_m$ be m subsets of a set of size n. Prove that

$$\sum_{i=1}^{m}\sum_{j=1}^{m}|A_i|\cdot|A_i\cap A_j| \geqslant \frac{1}{mn}\left(\sum_{i=1}^{m}|A_i|\right)^3.$$

Solution. WLOG assume $A_1, A_2, \ldots, A_m$ are subsets of $\{1, 2, \ldots, n\}$. Consider an $m \times n$ table. For the ith row, we colour its jth cell iff $j \in A_i$.

Let N be the number of triples (X, Y, Z) of coloured cells such that X and Y lie in the same row (probably $X = Y$), while Y and Z lie in the same column (probably $Y = Z$). Suppose X and Y lie in the ith row, while Z lies in the jth row. There are $|A_i \cap A_j|$ ways to choose the pair of coloured cells Y and Z lying in the same column (where the column index is an element in $A_i \cap A_j$), and $|A_i|$ ways to choose a coloured cell X. Thus, we have

$$N = \sum_{i=1}^{m} \sum_{j=1}^{m} |A_i| \cdot |A_i \cap A_j|.$$

For each $j = 1, 2, \ldots, n$, let s_j be the number of subsets containing j. Equivalently, it is the number of coloured cells in the jth column. By first choosing a coloured cell Y in the ith row and jth column, it is clear that

$$N = \sum_{j \in A_i} |A_i| \, s_j.$$

Also, it is easy to compute $\displaystyle\sum_{j \in A_i} \frac{1}{|A_i|} = m$ (for each i, there are $|A_i|$ copies of $\dfrac{1}{|A_i|}$) and $\displaystyle\sum_{j \in A_i} \frac{1}{s_j} = n$ (for each j, there are s_j copies of $\dfrac{1}{s_j}$). Therefore,

$$\begin{aligned}
&mn \sum_{i=1}^{m} \sum_{j=1}^{m} |A_i| \cdot |A_i \cap A_j| \\
&= \left(\sum_{j \in A_i} \frac{1}{|A_i|} \right) \left(\sum_{j \in A_i} \frac{1}{s_j} \right) \left(\sum_{j \in A_i} |A_i| \, s_j \right) \\
&\geqslant \left(\sum_{j \in A_i} 1 \right)^3 \qquad \text{(by Hölder's inequality)} \\
&= \left(\sum_{i=1}^{m} |A_i| \right)^3 .
\end{aligned}$$

This is the same as the desired inequality. $\square$